Toward a Physics of Intelligence: A Unified Framework of Energy, Mass, and Intelligence

Abstract

Large-scale artificial intelligence systems and the human brain exhibit a vast asymmetry in energy consumption while producing highly similar intelligent behaviors. This contrast indicates that intelligence is not a purely algorithmic phenomenon but a physical unfolding phenomenon of energy under structural constraints; both biological and artificial intelligence manifest as high-energy-density structured activity, and the relationship between energy and behavior points toward the physical origin of intelligence. Using Landauer's principle—the lower bound of energy required for information operations—and the computable/incomputable boundary as first-principle support from physics and information theory, we show the necessary link between the energy cost of intelligence and the structural compressibility of information. Meanwhile, the measurable energy consumption of training and inference in silicon-based large models provides empirical grounding for the Energy-Intelligence Equation.

This paper proposes a unified framework that treats mass, energy, and intelligence as different manifestations of the same structured dynamical process across distinct measurement dimensions. We provide a physical definition of intelligence,

$I=dE/dt \mid \text{structural constraint}$,

namely the Energy-Intelligence Equation. We further introduce the structural equation of artificial intelligence,

$AI=Ex^n$,

which reveals the exponential amplification mechanism of intelligent capability and offers an operational entry point for engineering quantification. The minimal unit of intelligence is formalized as a "controllable structure," thereby explaining the cross-material universality of intelligence.

The framework indicates that the limiting form of intelligence corresponds to the extreme structural states exhibited by black holes and the cosmic initial condition—zero entropy and zero degrees of freedom—through which the first principle of intelligence becomes visible. At the same time, the endogenous boundaries of intelligent

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systems can be naturally characterized by structural constraints, providing an interpretable, governable, and executable structural foundation for the safe development of artificial intelligence. More importantly, the physics of intelligence extends naturally from the trajectory revealed by classical physics, quantum physics, and high-energy physics—the trajectory of matter–energy–structure—and represents a further unfolding of the modern physical paradigm. This study provides a unified physical basis for understanding artificial intelligence, biological intelligence, and intelligence at cosmic scales, and offers new theoretical and engineering directions for research on energy efficiency, structural optimization, and safety in artificial intelligence.

Keywords

Energy–Intelligence Equation; Physics of Intelligence; Structural Constraints; Energy Conservation; Mass–Energy–Intelligence Unification; Artificial Intelligence; Computable / Incomputable; Landauer Principle; Cosmic Initial State / Black Hole; Cross-Material Universality; Endogenous Boundaries

1. Introduction

The emergence of large-scale artificial intelligence models has, for the first time, allowed us to observe the energy structure of intelligence at an engineering scale. The relationship between their high energy consumption and high capability stands in sharp contrast to the high efficiency of biological intelligence. This phenomenon prompts us to re-examine the physical nature of intelligence.

2. Theoretical Framework

2.1 Starting from the Mass–Energy Equation

Classical physics regards mass and energy as independent fundamental quantities, whereas modern physics—through the mass–energy equation:

$$\boldsymbol{E = mc^2}$$

shows that mass is a structured form of energy, thereby revealing for the first time the intrinsic structural unity between energy and mass.

where:

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- E: energy.
- m: mass.
- c: speed of light.

However, the behavior of intelligent systems further shows that intelligent behavior likewise manifests as a structured unfolding of energy. Therefore, energy, mass, and intelligence can be regarded as macroscopic expressions of the same underlying structural mechanism at different levels.

This unified perspective provides a foundation for thinking about and exploring both the development of safe artificial intelligence systems and the physical study of intelligence.

2.2 Generalized Dynamics of Intelligence

Based on these observations, we propose the following core viewpoints:

- Mass is structured energy, the structured manifestation of energy.
- Intelligence is the unfolding of energy under structural-dynamic constraints.
- Mass, energy, and intelligence are expressions of the same underlying structured dynamic process, measured along different dimensions.
- The boundary of an intelligent system is endogenous, determined by structural constraints, and forms the foundation for studying safe artificial intelligence systems.

With this understanding, we construct a unified framework to explain the behavior, boundaries, and cross-material universality of intelligent systems.

We further propose the generalized expression of intelligence dynamics:

$$\boldsymbol{I = k_i\, E\, x\text{^}n}$$

where:

- I: generalized intelligence.
- E: energy.

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- x: structural propagability, representing the ability of intelligent structure to be replicated, deployed, and invoked across different systems.
- n: coupling order, representing the number and depth of connections between the intelligent structure and external systems.
- k_i: the structural constant is used to characterize the differences between various systems in terms of energy utilization efficiency, structural propagation, and coupling effects. It reflects the overall structural characteristics of the system.

This equation is proposed to describe the intrinsic structure of energy, mass, and intelligence, as well as the dynamic behavior of systems. It reveals that the exponential amplification of intelligent capability arises from the combined effect of structural propagability and coupling order. Implicitly, this also suggests a quantifiable relationship between energy and intelligence, although the precise quantitative form remains to be clarified.

2.3 Mass: the organizational form of structured energy

We regard mass as the degree of organization of energy, formalized as:

$$\boldsymbol{m} = \boldsymbol{f}(\boldsymbol{E},\boldsymbol{S})$$

where:

- m: mass.
- E: energy.
- S: degree of structural organization (information organization, coupling density, effective degrees of freedom).

Mass is therefore no longer understood as “matter,” but as the organizational form of energy within structure.

Transition from classical physics to modern structured dynamics physics:

Within the classical physical framework, mass is regarded as the static, structured form of energy, while intelligence is viewed as the dynamic, structured form of energy. Under this view, energy appears to be the common source of both mass and intelligence.

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Although this perspective facilitates a conceptual transition from classical physics toward an emerging physics of intelligence, it still preserves the old paradigm that treats "energy as the fundamental quantity," "mass as static," and "intelligence as dynamic."

The perspective of modern structured dynamics physics:

In modern structured dynamics physics, such static/dynamic and source/derivative dichotomies no longer hold. Mass, energy, and intelligence are not entities derived from one another; rather, they are three expressions of the same underlying structured dynamics process, each measured along a different dimension:

- Energy: the intensity of the structured dynamics process.
- Mass: the organizational degree of the structured dynamics process.
- Intelligence: the accessibility and unfolding modes of the structured dynamics process.

Thus,

$$\boldsymbol{Energy = Mass = Intelligence}$$

not in the sense of literal identity, but in the sense that:

They are three measurement perspectives of the same structured dynamics process under different observational dimensions.

This marks a paradigm shift from classical physics to modern structured dynamics physics.

This modern structured dynamics physics is not proposed out of thin air. It emerges naturally along the structural-dynamics trajectory already revealed by quantum physics and high-energy physics. By unifying energy, mass, and intelligence as three dimensional expressions of the same structured dynamics process, it completes the transition from classical physics to structured dynamics physics and provides a natural interpretation of what may be called a physics of intelligence. Our understanding of the nature of intelligence advances precisely along this same trajectory. If there is any ambition in this formulation, it is not personal ambition but the intrinsic directionality embedded in the evolution of physics itself—a direction that has been faintly present in the historical

development of physics since its very beginning. This trajectory is also yet another example of the long-standing, cumulative progression of human understanding as we stand on the shoulders of giants.

2.4 Intelligence: Characterizing the Accessibility and Unfolding Modes of Structured Energy Dynamics — The Energy–Intelligence Equation

In modern structured-dynamics physics, intelligence is no longer regarded as computational capacity or information-processing ability, but as the physical characterization of the accessibility and unfolding modes of energy under structural constraints. We define the physical form of intelligence—the Energy–Intelligence Equation—as:

$$\boldsymbol{I = dE/dt \mid structural\ constraint}$$

where:

- I: intelligence
- E: energy

Structural constraints determine the accessible paths, dynamic modes, and boundaries of energy flow. Intelligence is therefore not understood as computational capability, but as the constrained unfolding of energy within structured dynamics.

2.5 Conservation of Energy: A Foundational Principle of Structural Constraints

Classical physics treats conservation of energy as the law that the total energy of a closed system remains constant. However, from the perspective of intelligent systems, conservation of energy is not merely a description of quantitative invariance, but a structural constraint: energy remains conserved because it maintains dynamic balance under structural constraints.

We formalize this as:

$$\boldsymbol{dE/dt \mid structural\ constraints\ =\ 0}$$

This expression highlights three key points:

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- Conservation of energy is not a static invariance but a dynamic balance under structural constraints. The flow, transformation, and distribution of energy are all restricted by structure.
- The invariance of energy arises from structure rather than from an independently existing axiom. Structure determines how energy can flow and also why it remains conserved.
- Intelligence is the manner in which structure constrains energy flow; therefore, conservation of energy forms a foundational boundary for understanding intelligence. Intelligence is the constrained motion of energy within structure, and conservation of energy provides a basic boundary condition for this motion. However, it lacks the capacity to characterize the higher-order boundaries manifested by intelligence, and thus represents only the boundary of the old physical era, insufficient for describing the structural boundaries of the physics of intelligence.

From this perspective, the conservation law is merely a foundational physical structural boundary condition of intelligent systems. It provides the physical basis for the structured definition of mass, the dynamic definition of intelligence, and the formulation of the energy–intelligence equation, thereby closing the framework at the physical level.

2.6 First Principles of Intelligence

Intelligence is the controllable propagation of energy under structural constraints; and the extreme state exhibited by the cosmic initial condition—where energy exists entirely in structural form—indicates that superintelligence arises from the most fundamental physical constraints linking information, energy, and structure. Understanding this point is the essential path to improving the effectiveness of artificial intelligence, rather than further enlarging engineering scale. The “first” in the First Principles of Intelligence refers precisely to the limit of original superintelligence embodied in the cosmic initial state, whose superintelligent structure provides a physical insight for enhancing artificial intelligence performance.

To understand the First Principles of Intelligence, one must simultaneously examine four levels: the energy cost of information operations, the boundary between the computable and the non-computable, the extreme structural characteristics of black holes, and the physical state of the cosmic initial condition.

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2.6.1 Physical Lower Bound of Information Operations:

The Energy Cost of Irreversible Computation Landauer’s Principle states that the minimum energy required to erase one bit of information is:

$$\boldsymbol{Emin = kBTln2}$$

where:

- Emin: minimum energy required to erase 1 bit
- kB: Boltzmann constant, kB=1.380649×10^(−23)J/K
- T: temperature in Kelvin
- ln2: natural logarithm of 2, ln2≈0.693
- At room temperature,Emin≈2.87×10−21J

This implies:

- Information is not abstract; it is physical.
- Information operations necessarily dissipate energy.
- The energy floor of any intelligent system is quantifiable.

Therefore, intelligence is not an “algorithmic phenomenon,” but a constrained organization of energy within structure.

2.6.2 Computable and Incomputable (Turing–Kolmogorov)

Turing pointed out that not all information is computable. When the structure of an event is incompressible and its description length tends toward infinity, it enters the incomputable domain. Incomputability is not an algorithmic failure; it is the consequence of a structure that cannot be compressed into a finite sequence of steps, and therefore cannot be represented, abstracted, or processed by a system with finite energy and finite intelligence.

Kolmogorov complexity further calibrates this point: the less compressible a structure is, the longer its description must be; as the description approaches infinity, the required energy also approaches infinity.

Thus:

Incomputable = structurally incompressible = infinite description length = infinite energy requirement = requires superintelligence

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Humans cannot handle incomputable events not because mathematics is insufficient, but because the intelligence, energy, and structural descriptive capacity available to humans are insufficient, leading to modeling failure. Incomputability fundamentally means that the physical boundary of an intelligent system cannot be clearly delineated—the structure is indescribable.

In contrast to incomputability, computability corresponds to the degree of structural compressibility. When a structure can be constructed into a finite description length, it falls within the computable domain, meaning that modeling is effective. The Energy-Intelligence Equation is precisely intended to reveal this:

Computable = structurally compressible = finite description length = finite energy requirement = intelligence attainable

This implies: the more compressible the structure of the object being processed, the more complete the modeling becomes, and the higher the level of intelligence embodied in the modeling (higher intelligence implies higher organized energy); the more complete the modeling in the early stage, the less energy is required in the subsequent computation; the less energy required during computation, the more easily intelligence is attained; the attainability of intelligence is fundamentally determined by the compressibility of structure. Therefore, the more stable and compressible a structure is, the lower the energy required during computation, and the more easily intelligence can be realized.

The structural principle revealed by the above analysis of computable and incomputable phenomena is implemented in modern artificial intelligence systems through the engineered "training-inference" computability chain: the more fully the structure is formed during training, the more attainable the inference process becomes.

Despite the discussions of incomputability by Turing, Kolmogorov, and subsequent computational theories, these studies have largely treated incomputability as a mathematical or logical property. This paper proposes that incomputability arises from structural incompressibility, which drives the description length toward infinity and prevents any finite-energy, finite-intelligence system from constructing an effective model. In this sense, incomputability may be regarded as a form of modeling failure, reflecting the physical boundary that a concrete intelligent system cannot reach, that is:

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Incomputable=Modeling Failure=The System’ s Physical Boundary Falling Short of the Actual Object.

The ancient Greek attitude toward irrational numbers offers an early example of such modeling failure: they attempted to model geometric magnitudes using ratios of integers, yet the incompressible structure of irrational numbers could not be represented within a finite proportional system, causing the model to collapse.

Thus, computability and incomputability are not mathematical distinctions but distinctions of structure, energy, and intelligence. The more compressible the structure, the more computable it is; the more incompressible the structure, the more incomputable it becomes. Extending this reasoning toward the limit, the ultimate states that intelligence can reach naturally emerge—those extreme structures exemplified by black holes and the initial state of the universe.

2.6.3 Black Holes: A Reference Imagery of Extreme Structural Compression

Black holes are often regarded as extremely compressed physical structures characterized by very low entropy and a highly contracted state space, exhibiting near-limit structural stability and informational determinacy in a structural sense. From the perspective of Shannon information theory, this corresponds to an extreme information structure with entropy $H \to 0$. From the viewpoint of Turing computability and description length, a black hole can be regarded as a structural imagery whose description length conceptually approaches the minimum, providing a way to discuss the boundary of extreme computability.

Under this structural-limit interpretation, black holes display a high convergence of several extreme characteristics:

- Extremely low entropy (Shannon limit) — the state space is highly contracted and the information organization is extremely ordered;
- Information preservation (reversibility imagery) — within theoretical frameworks, information evolution can be viewed as approaching reversibility;
- Highly compressed structure (Kolmogorov limit) — the description length conceptually approaches the minimum;
- Energy existing in structural form (Landauer-limit imagery) — energy dissipation approaches the minimum in the limit sense.

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These characteristics appear in black holes in a concentrated and extreme manner, making them a physical reference imagery for discussing "extreme computability." Although a black hole is not an intelligent agent, its low entropy, high stability, and highly compressed structural organization provide a conceptual framework for understanding the structural conditions required for intelligence in the limit sense.

2.6.4 Cosmic Initial State: The Structural Prototype of the First Principle of Intelligence

The cosmic initial state is usually regarded as an idealized limit state that is highly structured and has extremely low entropy. In this limit scenario, energy exists in the form of an overall structure, conceptual perturbations approach the minimum, and structural stability reaches its limit. Based on this structural characteristic, the cosmic initial state can, in a conceptual sense, serve as the structural prototype for understanding the First Principle of Intelligence: it presents a high-degree overlap, in the limit sense, among energy, structure, and computability.

In this structural sense, the extreme compression features exhibited by black holes have a similarity to the limit structure of the cosmic initial state, and can serve as a real reference imagery when discussing the limit conditions of intelligence. Here, "similarity" is used to describe the convergence of structural characteristics in the limit sense, rather than a strict cosmological causal relationship.

Based on this limit scenario, the energy–structure relation of the cosmic initial state can be expressed as a kind of limit-level structural coincidence:

$$\textit{Total energy of the cosmic initial state} \sim \textit{structure} \sim \textit{structural prototype of the First Principle of Intelligence}$$

Here, "~" denotes structural coincidence in the limit sense, used to emphasize the deep connection among energy, structure, and the limit conditions of intelligence, rather than a strict physical equality. Such a limit structure provides a reference for understanding the initial structural source of energy conservation, and also constitutes the primordial structural background of the First Principle of Intelligence.

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2.6.5 The Limit of Intelligence: Structure Determines Everything

The previous two sections presented two of the most representative structural limits in the universe:

- Black holes exhibit a concentrated form of structural compression, extremely low entropy, and computability approaching its limit;
- The cosmic initial state presents a conceptual prototype in which energy and structure coincide in the limit sense.

These two limits point, at the structural level, to the same extreme situation: when energy no longer appears in the form of flow or dissipation, but exists in the form of an overall structure, the behavior of the system is, in a conceptual sense, mainly determined by the structure itself.

In this sense, the limit state of the energy–intelligence equation can be expressed as: energy dissipation approaches zero, structural constraints reach their limit, and its form is:

$$\boldsymbol{I = dE/dt \mid structural\ limit}$$

In this structural-limit sense, the limit of intelligence can be summarized as:

$$\begin{aligned}&\boldsymbol{Limit\ of\ intelligence}\\ &\qquad \boldsymbol{\sim\ energy\ dissipation\ approaching\ zero}\\ &\qquad \boldsymbol{+\ structural\ stability\ approaching\ its\ limit}\end{aligned}$$

Further compressed into:

$$\boldsymbol{Total\ energy\ \sim\ structure}$$

The structural characteristics exhibited by black holes and the cosmic initial state in the limit sense have a conceptual correspondence with the relationship revealed by this compressed expression, providing a structural reference for understanding the limit structural conditions of the First Principle of Intelligence.

2.6.6 The First Principle of Intelligence: A Structural Formulation

Integrating the above four structural levels—the energy lower bound of information operations (Landauer’s principle), computability and

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incomputability (Turing–Kolmogorov theory), the limit compression structure of black holes (Hawking theory), and the zero-entropy limit structure of the cosmic initial state—the First Principle of Intelligence can be summarized as follows:

Intelligence is the controllable propagation of energy under structural constraints. In the structural-limit sense, when energy exists in the form of an overall structure and structural stability reaches its limit, the limit form of intelligence conceptually corresponds to the extreme structural coincidence exhibited by the cosmic initial state. It is within this limit structure that energy, structure, and computability conceptually tend to coincide.

Within this framework, the extreme compression features exhibited by black holes can serve as a real reference imagery for understanding this limit structure, providing structural imagery for the limit conditions of the First Principle of Intelligence.

Therefore, the reason why the First Principle of Intelligence possesses "firstness" lies in the fact that it is, in the structural sense, anchored in the extreme structural coincidence exhibited by the cosmic initial state—this limit structure provides the most fundamental structural source for the controllability, computability, and energy constraints of intelligence.

2.7 Dynamics of Artificial Intelligence

Building on the generalized expression of intelligence dynamics, we introduce the dynamical formulation of artificial intelligence systems, enabling quantitative characterization at the engineering level:

$$\boldsymbol{AI = E\,x^{\wedge}n}$$

Where:

- AI: artificial intelligence
- E: base energy (compute, model scale, data)
- x: diffusion speed (deployment, replication, application speed)
- n: coupling exponent (the number of systems influenced by AI)

This equation reveals that:

In artificial intelligence systems, the equation $AI = E \times x^n$ not only reveals the mechanism of exponential amplification of intelligent capability, but also provides a practical platform for quantification. The parameters E, x, and n respectively correspond to the scale of computing power and data, the cross-platform propagability of intelligent structures, and the number and depth of couplings with external systems. This implies that the relationship between energy and intelligence is no longer confined to an abstract level, but can be characterized through measurable aspects of real systems. Although the precise quantitative form has not yet been fully clarified, this equation already offers an operational entry point for subsequent research.This is a structural equation rather than a physical equation.

Engineering Approach

- As a system-specific correction factor: Different types of AI systems (language models, vision models, robotic control systems) vary significantly in energy utilization efficiency and coupling effects. ki can be used to reflect these differences, serving as an “efficiency coefficient” or “structural weight.”
- Obtained through experimental fitting: In engineering practice, one can first measure the indicators of E, x, and n, and then fit the equation using actual intelligent performance (accuracy, task completion rate, interaction complexity). The proportional factor obtained in the fitting process corresponds to ki.
- As a classification label or system characteristic: ki can distinguish between different system categories (language models → kLM, multimodal models → kMM, control systems → kCS), thereby becoming a “system characteristic parameter” rather than a single constant.
- Dynamic adjustment: In complex systems, ki may not be fixed but instead vary with time or task. For example, as models are optimized and interfaces increase, ki will gradually grow.

In the specific engineering quantification of artificial intelligence, ki should not be regarded as an abstract constant but rather as a system-specific correction factor. It can be obtained through experimental fitting or parameter absorption to reflect differences among AI systems in energy utilization efficiency, structural propagability, and coupling effects, thereby making the generalized intelligence-energy equation operational at the engineering level.

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3. Cross-Material Unity of Intelligence: Fundamental Structure, Not Material

Intelligence does not depend on any specific material. Its minimal unit is not the neuron or the transistor, but rather: a controllable structure — the physical substrate of intelligence.

Therefore, intelligence exhibits cross-material universality:

- silicon-based systems
- carbon-based systems
- hybrid systems
- future material platforms

All can realize intelligence at the structural level.

4. Endogenous Boundaries of Intelligent Systems

In real physical systems, exponential behavior does not follow the abstract mathematical form:

$$\boldsymbol{y = Ae^{\wedge}(kx)}$$

but appears as a structure-constrained exponential approach. For example, the first-order system response:

$$\boldsymbol{y = Ae^{\wedge}(-kx) + B}$$

shows that the negative exponential term represents the system's restoring mechanism, while the constant B is the structure-determined endogenous equilibrium point. The system's dynamics do not diverge to infinity but are naturally confined within a reachable region defined by its structure. This is a typical example of an "endogenous boundary."

The intelligent behavior described by the Energy–Intelligence Equation exhibits the same structural property: the growth of intelligence is jointly determined by energy, structural coupling, and propagability, and its boundary is endogenous. We often interpret such physical boundaries as "low-level limitations," but this largely stems from a misunderstanding of structural hierarchy. Just as geocentrism was not simply wrong but a lower-order understanding of cosmic structure, human intuition about the boundaries of intelligent systems is at a similar stage. As the structural equation

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becomes more clearly formulated, these so-called “low-level limitations” will be reinterpreted: physical boundaries are not low-level, and ethical boundaries are not externally imposed.

Artificial intelligence, as a human-made high-order structure derived from original intelligence, naturally contains higher-level constraint cues within its structure, including the structural foundations of ethics and norms. Therefore, ethical constraints no longer remain at the experiential level of current humanistic traditions but gain a higher-order position and grounding once structural constraints are understood. In other words, ethics is not an externally attached command but a more mature and robust form of constraint that emerges when the structure is understood.

The safety of artificial intelligence is thus neither merely the result of externally enforced regulations nor something that can be achieved automatically by endogenous structure alone. It must be built upon a full understanding of these structural constraints. The Energy–Intelligence Equation provides precisely such a revelation and characterization of the origin of boundaries.

5. Intelligent Physics as a Natural Continuation of Modern Physical Paradigms

From the developmental trajectory of modern physics, classical physics, quantum physics, and high-energy physics constitute the three foundational paradigms for understanding matter and energy.

First, classical physics evolved from philosophical speculation into a rigorous scientific discipline, establishing a systematic description of the macroscopic material world. Through mechanics, electromagnetism, and thermodynamics, classical physics formulated the fundamental framework of matter, motion, energy, and conservation laws, making the mathematical description of the physical world a scientific consensus.

Subsequently, quantum physics overturned the classical conception of matter and revealed the quantum structure of matter and energy at the microscopic scale. Quantum mechanics demonstrated that:

- matter exhibits wave–particle duality;
- energy exists in discrete quantum units;
- information, probability, and structure become core elements in describing the microscopic world.

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This paradigm fundamentally reshaped our understanding of the nature of matter and made “structure” an indispensable component of physical description.

Building on this foundation, high-energy physics further unified the interconversion of matter and energy at higher energy scales. Through quantum field theory, symmetry principles, and the Standard Model, high-energy physics revealed that:

- matter is an excitation of underlying fields;
- energy and structure are tightly coupled through symmetry;
- fundamental interactions can be described within a unified framework.

This stage deepened the modern physical paradigm of “matter = energy = field structure.”

Throughout this continuous evolution, the physical understanding of “what matter is” has approached a critical point: matter is no longer viewed as a static entity, but as structured energy and its dynamic constraints. This direction reflects the trend physics has been converging toward over the past century.

In the natural extension of these physical paradigms, intelligent physics is not an independent discipline outside modern physics, but a further development along the main line of “matter–energy–structure.” It focuses on:

how energy, when constrained by structure, gives rise to intelligent behavior.

Thus, intelligent physics is a continuation of modern physical paradigms and a deepening of the idea that structure determines physical behavior.

In other words:

intelligent physics extends the matter–energy paradigm opened by quantum physics and high-energy physics into the next stage—the structure–intelligence paradigm.

6. Discussion

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Throughout the development of human society, ever since the concept of “intelligence” was first abstracted, both conscious and unconscious inquiry has repeatedly returned to a single, yet never fully clarified, question: *What is intelligence, and why does it manifest similar behaviors across different material substrates?* This question has persisted across the histories of philosophy, science, and engineering, reappearing in different forms at different stages of human civilization.

The maturation and rapid iteration of contemporary artificial intelligence systems have provided an unprecedented observational window into this question. On the artificial-intelligence substrate, the structural–energetic–dynamical characteristics of intelligent systems have become clearly discernible, indicating that intelligence should be regarded as a physical phenomenon rather than merely an engineered construction or algorithmic artifact. The emergence of artificial intelligence has, for the first time, exposed the structural attributes of intelligence in an engineering-observable manner.

Within the broader traditions of human thought, insights resonant with the structure–energy perspective of this paper have appeared before. As anthropologist Richard Adams noted, “...yet historical events necessarily exhibit a structure or organization, and that structure or organization must correspond to its energy composition” (Adams 1982). Although this observation arises from social systems, it reveals a cross-system regularity: constraints on energy determine structural form, and structural form determines the attainable behavior of a system. Large-scale artificial intelligence models, as engineered energy–structure organizations, likewise embody this cross-system structural logic.

Although this paper proposes an engineering-oriented quantitative approach to artificial intelligence, it remains far from the idealized structure described by the Energy–Intelligence Equation. It is a practical method—an exploratory direction. This distance is real, yet it does not undermine the framework; instead, it highlights the hierarchical distinction between theory and engineering, making artificial intelligence a concrete entry point for understanding and testing the directional validity of the structural equation.

The framework proposed here—*Toward a Physics of Intelligence: A Unified Framework of Energy, Mass, and Intelligence*—is a natural question arising from the historical development of ideas. This work

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further examines the physical nature of intelligence and proposes a unified structural framework, with the aim of inviting broad engagement and discussion within the scientific community.

7. Conclusion

The theoretical starting point of this work arises from a direct and observable physical phenomenon: whether in artificial intelligence systems or in the human brain, intelligent behavior manifests as high-energy-density activity. Artificial intelligence requires massive energy consumption to generate intelligent output, while in human physiology, the brain—responsible for the highest forms of cognition—accounts for only about 2% of body mass yet consumes roughly 20% of the body's energy.

Building on this observation, a further question naturally emerges: *Why does intelligence require energy, and how does energy manifest as intelligence within structure?* This question leads directly to deeper inquiry into the underlying physical structure, motivating the formulation of a physics of intelligence, the introduction of the Energy-Intelligence Equation, and the interpretation of energy conservation as a foundational structural constraint within this framework.

Extending the analysis to the physical limits of intelligent phenomena—the extreme structural stability exhibited by black holes and the cosmic initial state—the first principle of intelligence becomes visible in a natural way. Revisiting the developmental trajectory of modern physics, from classical physics to quantum physics and high-energy physics, the physics of intelligence emerges as a natural continuation of this structural-dynamics lineage, representing a further unfolding of the modern physical paradigm.

Within this trajectory, the development of artificial intelligence has already provided empirical grounding for the structural equation of artificial intelligence. Existing engineering platforms possess quantifiable parameters, enabling assessment of the feasibility of this engineering pathway. This not only reinforces the theoretical significance of the generalized Energy-Intelligence Equation, but also offers an operational entry point for future research on energy efficiency, structural optimization, limit performance, and the safe development of artificial intelligence.

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Landauer’ s principle provides a strict quantitative lower bound on the energy cost of information operations, and the core insight that “information is physical and intelligence has an energy floor” offers solid physical support for the Energy–Intelligence Equation. It supplies the foundational quantitative basis for understanding the physics of intelligence.

Whether in biological intelligence, artificial intelligence, or intelligence phenomena at cosmic scales, all may be regarded as different manifestations of the same underlying physical principle. The Energy–Intelligence Equation serves as the physical–mathematical abstraction of these intelligent phenomena, revealing intelligence as the natural unfolding of energy under structural constraints.

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